1 Two identical beakers, each of mass M, are attached to each other using string and suspended from a pulley, as shown in Fig. 1.1.

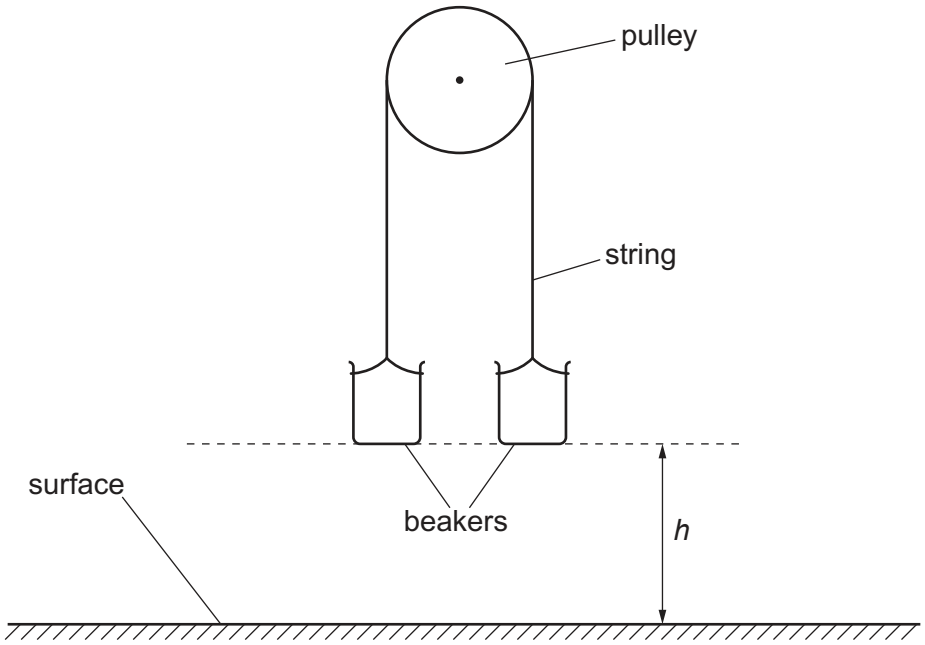

Fig. 1.1 (not to scale)

The beakers are held at rest at a height h above a surface. Cooking oil of volume V is added to one of the beakers.

The beakers are released so that the beaker with the oil begins to fall. The speed of the beaker as it reaches the surface is z.

It is suggested that z is related to V by the relationship

$$\frac{2h}{z^2} = \frac{2M}{abV} + \frac{1}{b}$$

where a and b are constants.

Plan a laboratory experiment to test the relationship between z and V.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for a and b.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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